

Matrix Algebra

Geometric Algorithms Lecture 9

Recap: One-to-one and Onto

Keywords

one-to-one transformation

onto transformation

matrix multiplication

row-column rule

matrix addition and scaling

non-commutativity

Onto and One-to-One

Onto and One-to-One

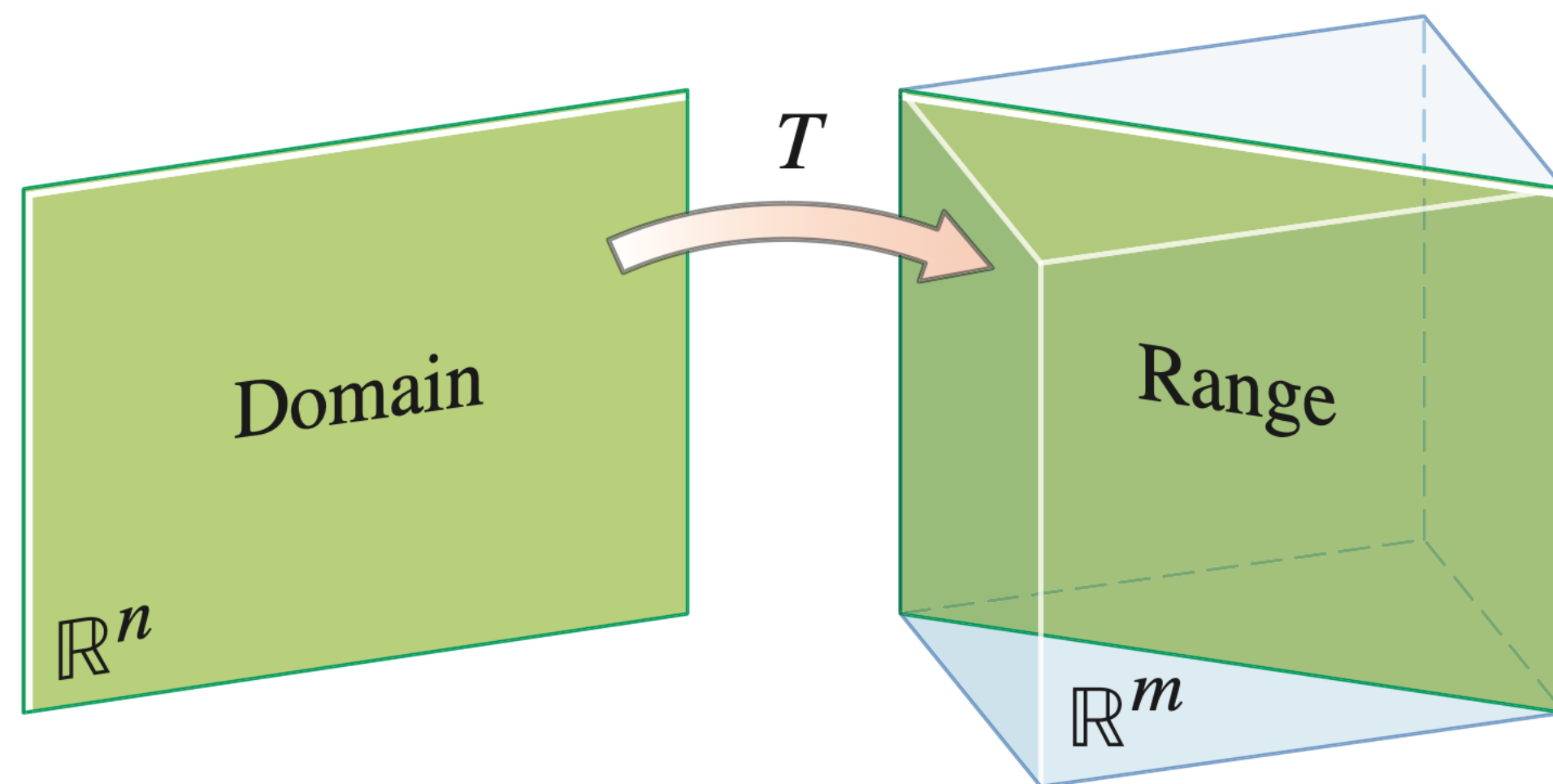
Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *onto* if any $\mathbf{b} \in \mathbb{R}^m$ is the image of at least one vector $\mathbf{v} \in \mathbb{R}^n$ (i.e., $T(\mathbf{v}) = \mathbf{b}$)

Onto and One-to-One

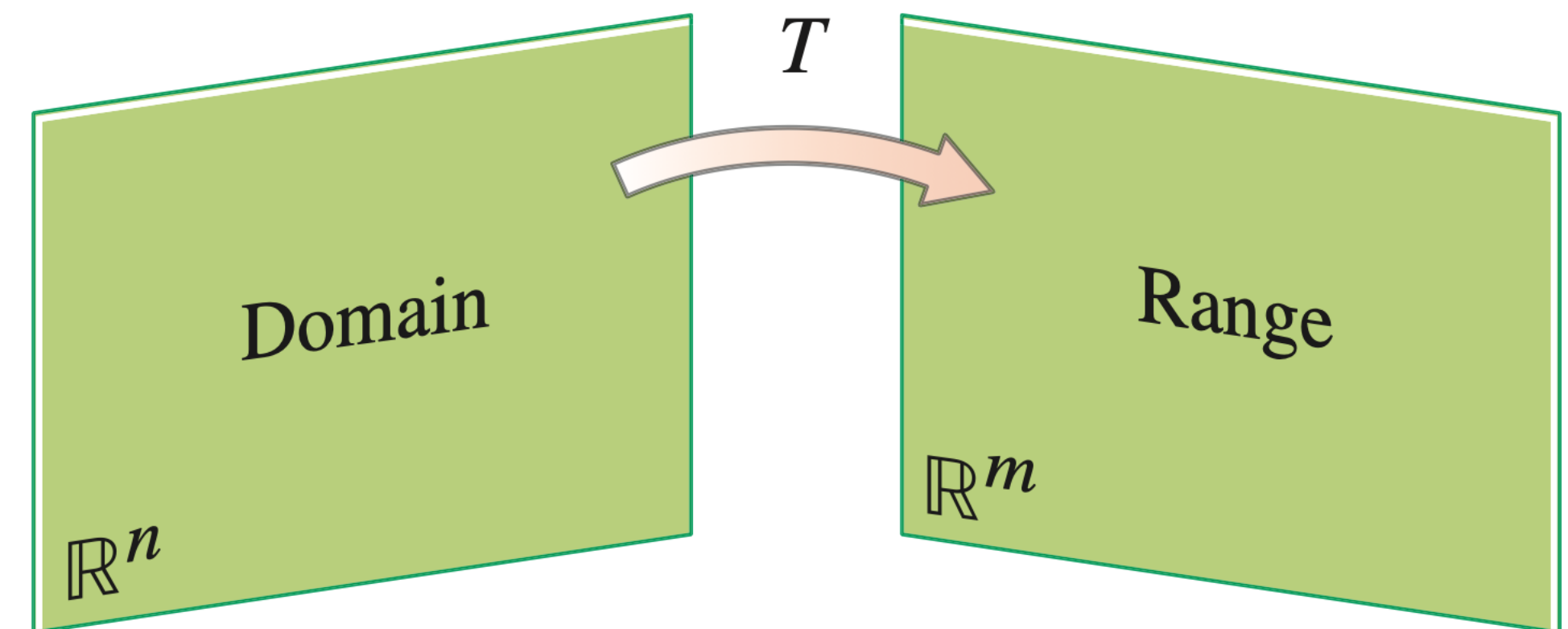
Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **onto** if any $\mathbf{b} \in \mathbb{R}^m$ is the **image of at least one vector** $\mathbf{v} \in \mathbb{R}^n$ (i.e., $T(\mathbf{v}) = \mathbf{b}$)

Df. $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is **one-to-one** if any $\mathbf{b} \in \mathbb{R}^m$ is the **image of at most one vector** $\mathbf{v} \in \mathbb{R}^n$

Onto (Pictorially)

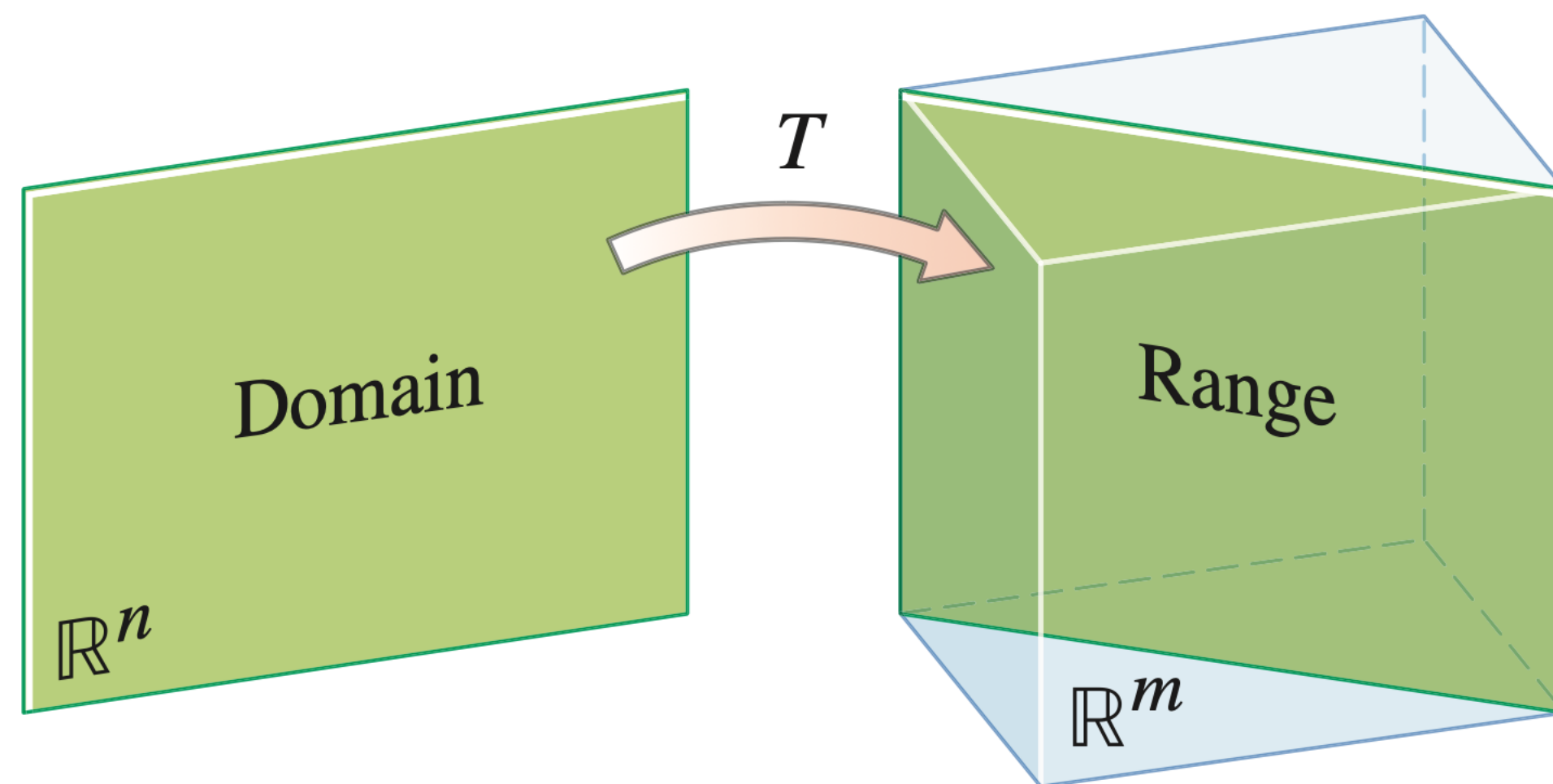


T is *not* onto $\mathbb{R}^m$

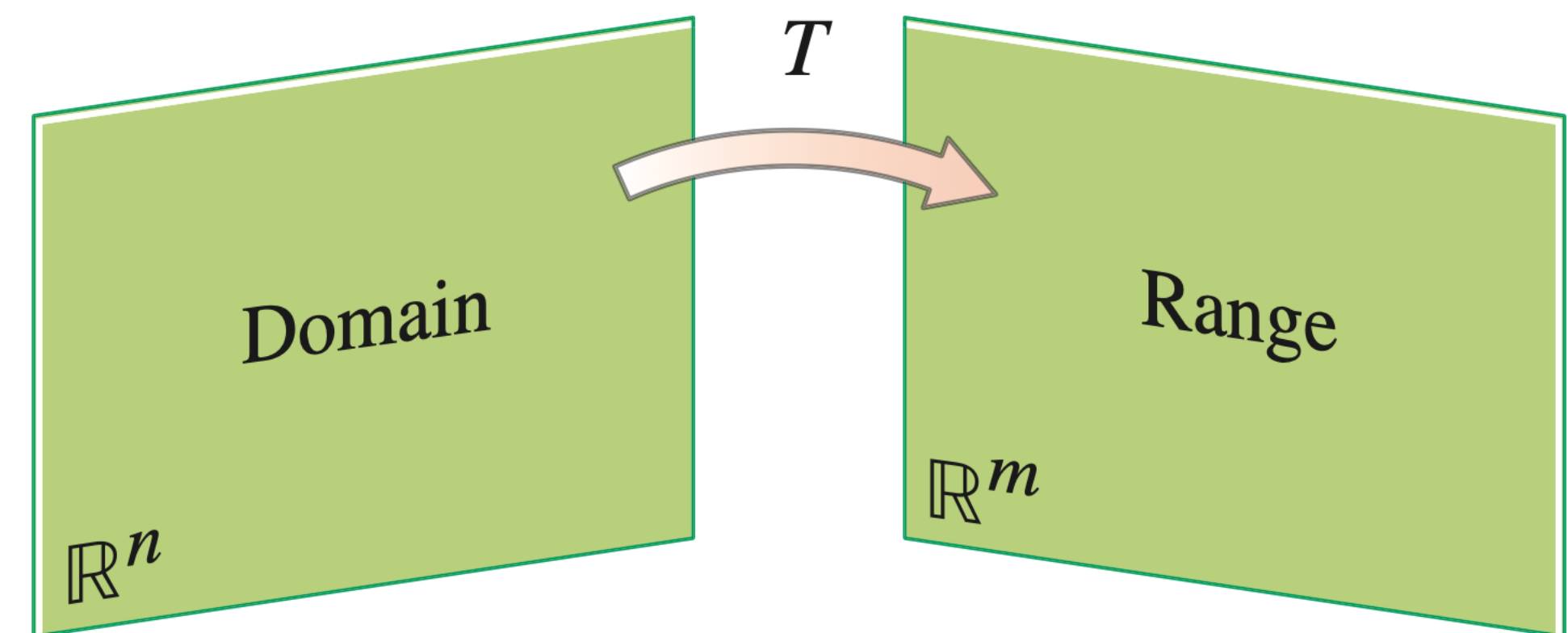


T is onto $\mathbb{R}^m$

Onto (Pictorially)



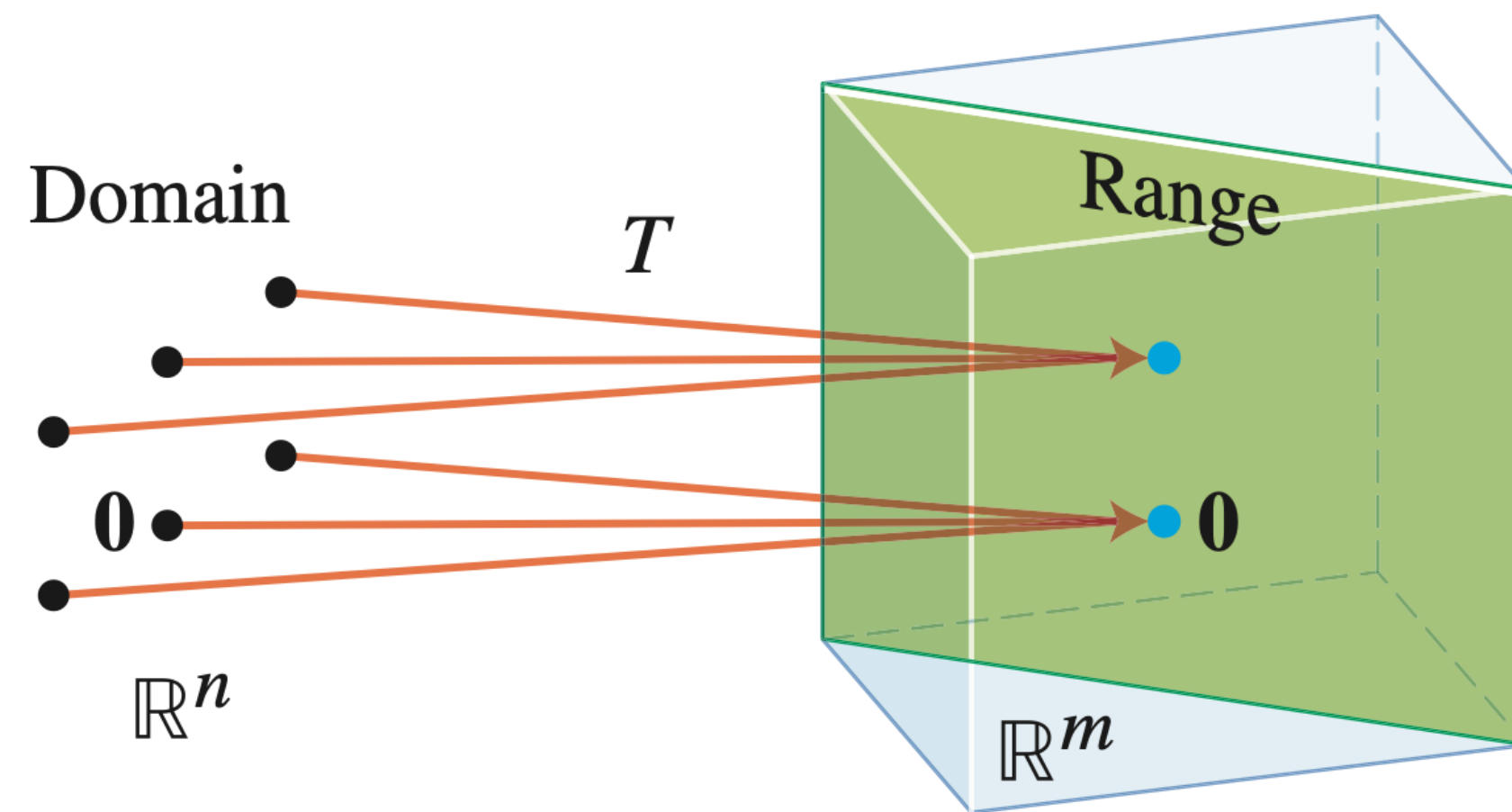
T is *not* onto $\mathbb{R}^m$



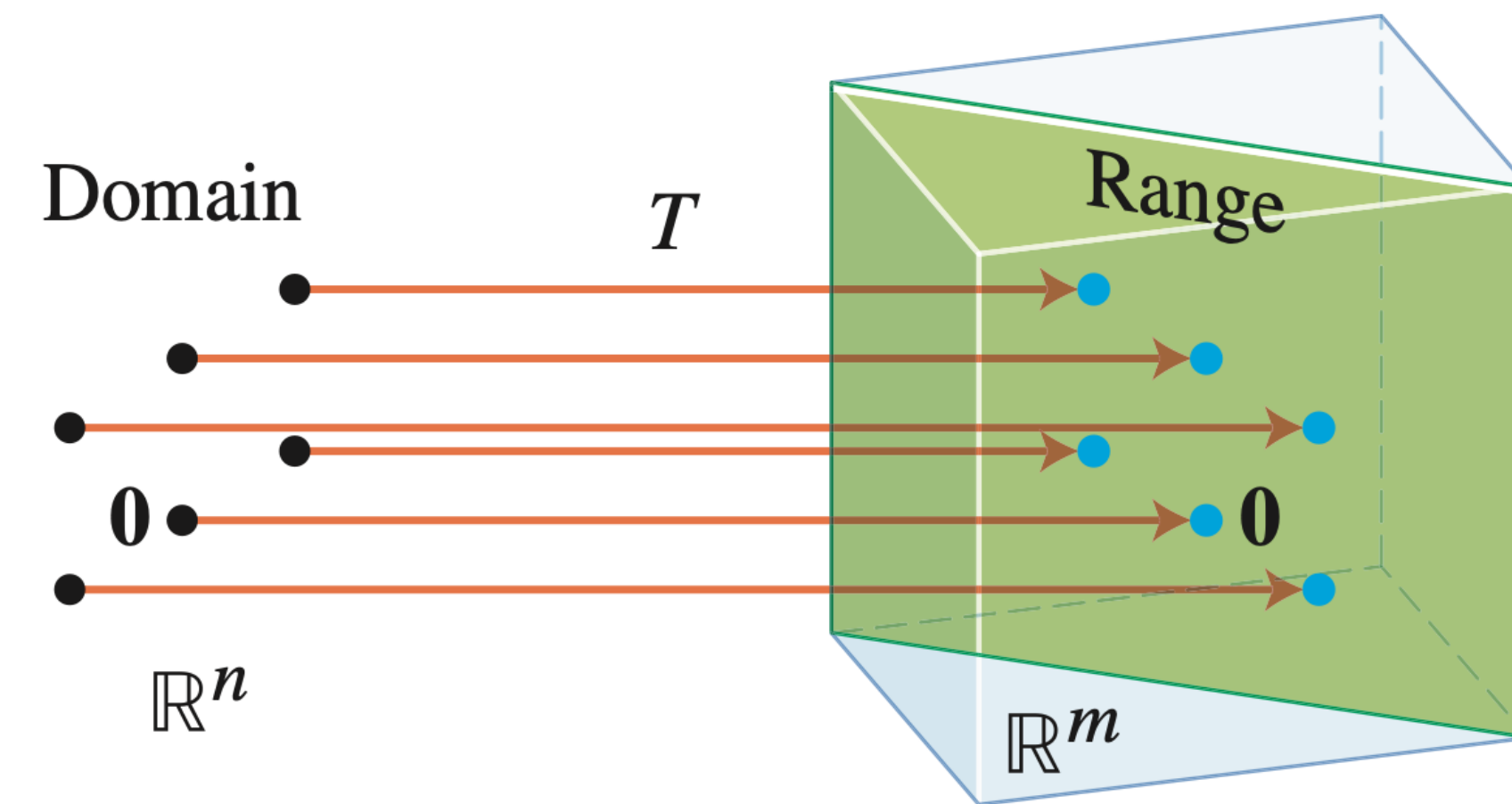
T is onto $\mathbb{R}^m$

T is onto if its range = its codomain

One-to-One (Pictorially)



T is *not* one-to-one



T is one-to-one

Taking Stock: Onto

Taking Stock: Onto

Thm. The following are logically equivalent for
 $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

Taking Stock: Onto

Thm. The following are logically equivalent for
 $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is onto

Taking Stock: Onto

Thm. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is onto

» $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$

Taking Stock: Onto

Thm. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is onto
- » $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$
- » $\text{range}(T) = \text{codomain}(T)$

Taking Stock: Onto

Thm. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is onto
- » $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$
- » $\text{range}(T) = \text{codomain}(T)$
- » the columns of A span $\mathbb{R}^m$

Taking Stock: Onto

Thm. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is onto
- » $A\mathbf{x} = \mathbf{b}$ has a solution for any choice of $\mathbf{b}$
- » $\text{range}(T) = \text{codomain}(T)$
- » the columns of A span $\mathbb{R}^m$
- » A has a pivot position in every row

Taking Stock: One-to-One

Taking Stock: One-to-One

Theorem. The following are logically equivalent
for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

Taking Stock: One-to-One

Theorem. The following are logically equivalent
for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is one-to-one

Taking Stock: One-to-One

Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

» T is one-to-one

» $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$

Taking Stock: One-to-One

Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is one-to-one
- » $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
- » $A\mathbf{x} = \mathbf{0}$ has only the trivial solution

Taking Stock: One-to-One

Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

- » T is one-to-one
- » $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
- » $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
- » The columns of A are linearly independent

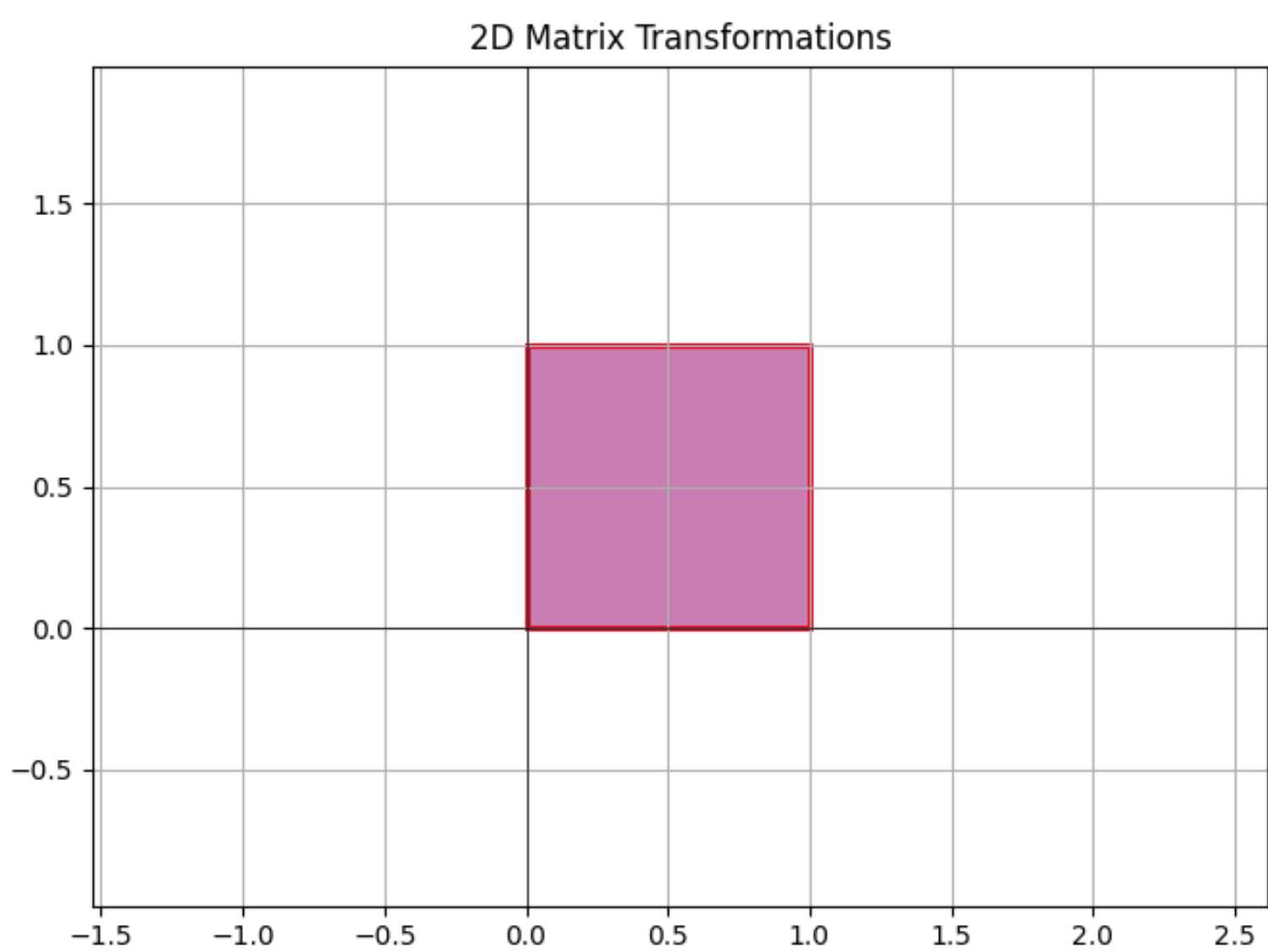
Taking Stock: One-to-One

Theorem. The following are logically equivalent for $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ implemented by A

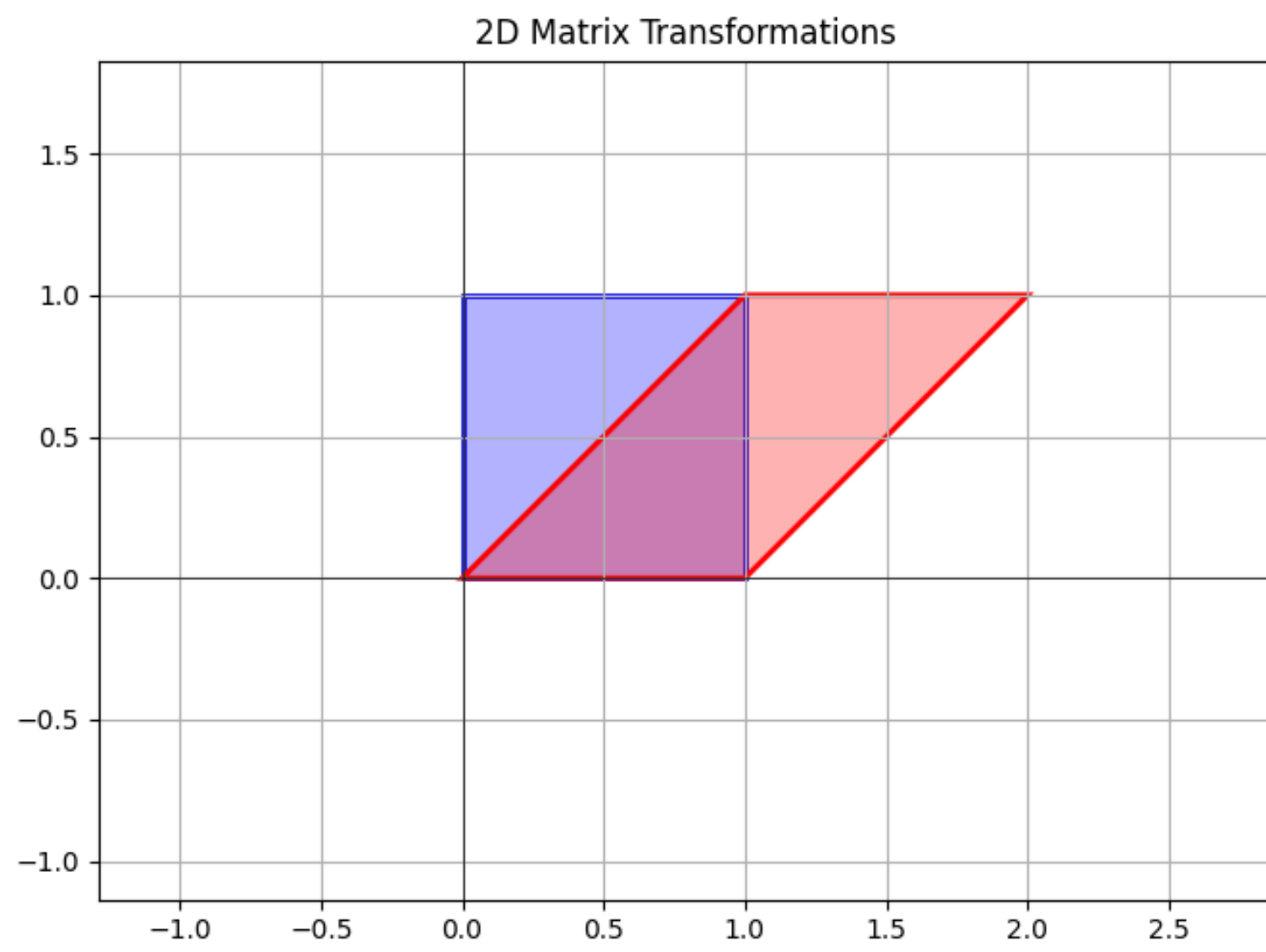
- » T is one-to-one
- » $A\mathbf{x} = \mathbf{b}$ has at most one solution for any $\mathbf{b}$
- » $A\mathbf{x} = \mathbf{0}$ has only the trivial solution
- » The columns of A are linearly independent
- » A has a pivot position in every column

Matrix Multiplication

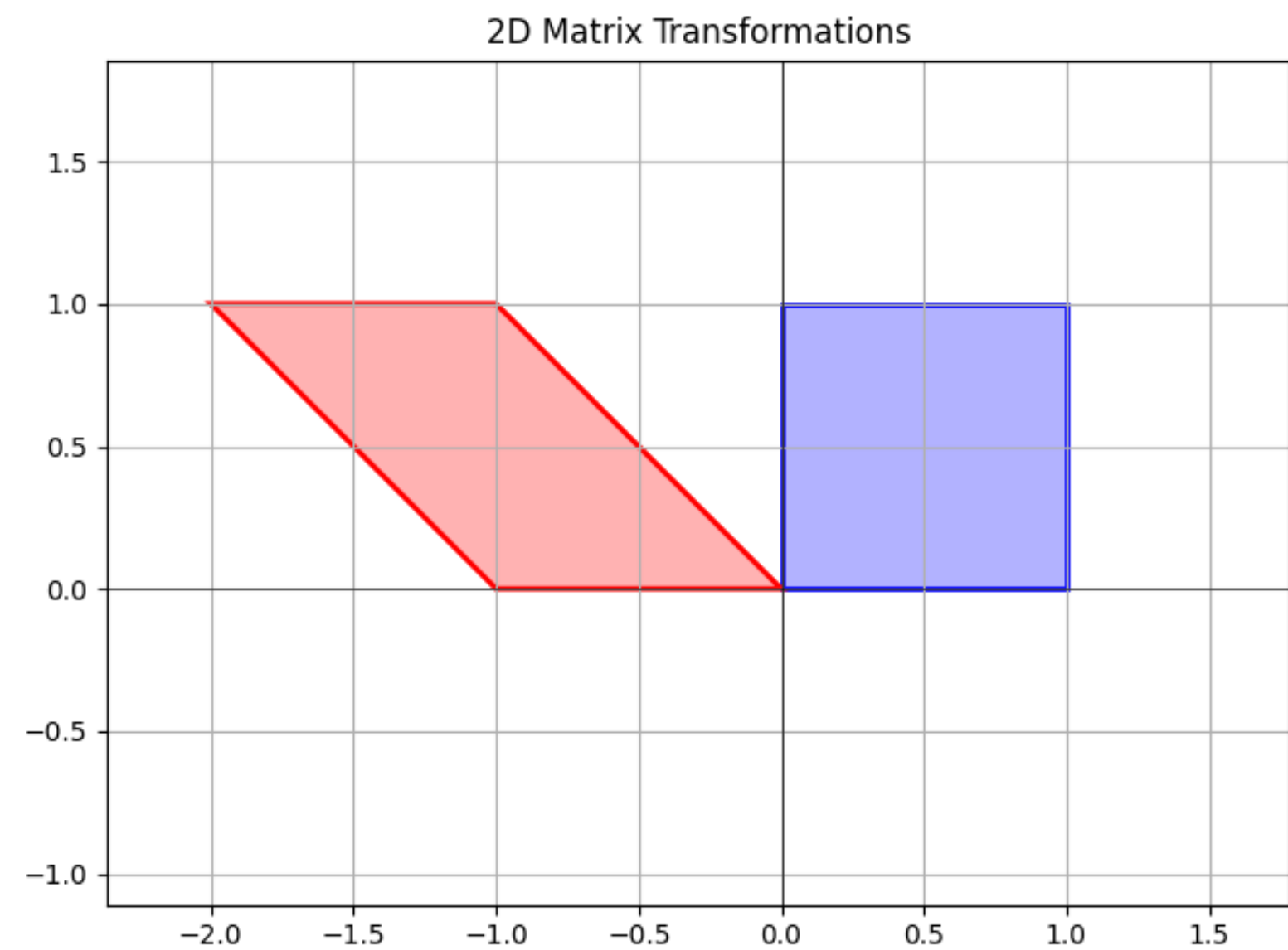
Shearing and Reflecting (Geometrically)



shear



reflect



Shearing and Reflecting (Algebraically)

$$\begin{matrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \end{matrix} \left(\begin{matrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{matrix} \right)$$

reflect shear

First multiply by shear matrix, then multiply
by reflection matrix

The Key Fact

The Key Fact

Thm. The composition of two linear transformations is a linear transformation

The Key Fact

Thm. The composition of two linear transformations is a linear transformation

This means the composition of two matrix transformations can be represented as a *single* matrix

The Key Question

*Given two linear transformations,
how to we compute the matrix which
implements their composition?*

Matrix Multiplication

Shearing and Reflecting

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) =$$

Matrix Multiplication

Df. For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ with columns $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_p$ the product $AB \in \mathbb{R}^{m \times p}$ is given by

$$AB = A \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \dots & \mathbf{b}_p \end{bmatrix} = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \dots & A\mathbf{b}_p \end{bmatrix}$$

Replace each column of B with A multiplied by that column

Shearing and Reflecting

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Tracking Dimensions

This only works if the number of columns of the left matrix matches the number of rows of the right matrix

The diagram illustrates the compatibility of matrix dimensions for multiplication. It shows three matrices with their dimensions explicitly labeled:

- Left Matrix:** A 5x3 matrix. The number of rows is labeled m (blue) and the number of columns is labeled n (red). Below it, the dimension is given as $(m \times n)$, with m in a blue box and n in a red box.
- Middle Matrix:** A 3x4 matrix. The number of rows is labeled n (red) and the number of columns is labeled k (purple). Below it, the dimension is given as $(n \times k)$, with n in a red box and k in a purple box.
- Result Matrix:** A 5x4 matrix. The number of rows is labeled m (blue) and the number of columns is labeled k (purple). Below it, the dimension is given as $(m \times k)$, with m in a blue box and k in a purple box.

The matrices are represented by grids of asterisks (*). The first matrix has 5 rows and 3 columns. The second matrix has 3 rows and 4 columns. The result matrix has 5 rows and 4 columns. The equality is indicated by an equals sign between the second and third matrices.

Important

Even if AB is defined, it may
be that BA is *not* defined

Non-Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \left[\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right]$$

Non-Example

The diagram illustrates a non-example of matrix multiplication. On the left, a 2x3 matrix (blue) is multiplied by a 2x2 matrix (green). This is followed by an equals sign and two possible ways to group the matrices, both of which are crossed out with large red 'X' marks. The first crossed-out expression shows the 2x3 matrix multiplied by the first column of the 2x2 matrix (a 2x1 vector). The second crossed-out expression shows the 2x3 matrix multiplied by the second column of the 2x2 matrix (another 2x1 vector). Both resulting expressions are enclosed in large square brackets, indicating they are not valid matrix products.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \left[\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} \right]$$

These are not defined

Example

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = \begin{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} \quad \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} \end{bmatrix}$$

The Key Fact (Restated)

For any matrices A and B such that AB is defined and any vector $\mathbf{v}$

$$A(B\mathbf{v}) = (AB)\mathbf{v}$$

The matrix implementing the composition is the product of the two underlying matrices

Row-Column Rule

Given $A \in \mathbb{R}^{m \times n}$ and $n \times p$ matrix $B \in \mathbb{R}^{n \times p}$ the entry $(AB)_{ij}$ is given as:

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with its first row highlighted in light blue. The second matrix is a 3x4 matrix with its first column highlighted in light red. An equals sign follows, and then a 5x4 matrix where the element in the first row and first column is highlighted in light purple, representing the result of the dot product of the first row of the first matrix and the first column of the second matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with its first row highlighted in light blue. The second matrix is a 3x4 matrix with its fourth column highlighted in light red. An equals sign follows, and then a 5x4 matrix where the element in the first row and fourth column is highlighted in light purple, representing the result of the dot product of the first row of the first matrix and the fourth column of the second matrix.

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

The diagram illustrates the Row-Column Rule for matrix multiplication. It shows three matrices: a 5x3 matrix A , a 3x4 matrix B , and their product AB , which is a 5x4 matrix. The second row of A is highlighted in light blue, the first column of B is highlighted in light red, and the element in the second row, first column of AB is highlighted in light purple. This visualizes the calculation of the element $(AB)_{21}$ as the dot product of the second row of A and the first column of B .

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements represented by asterisks (*). The second matrix is a 3x4 matrix, also with all elements represented by asterisks (*). The third matrix is a 5x4 matrix, also with all elements represented by asterisks (*). The second matrix is positioned between the first and third matrices, with an equals sign (=) to its right. The first matrix has its second row highlighted in light blue. The second matrix has its second column highlighted in light red. The third matrix has its second column highlighted in light purple. This visualizes the calculation of the element in the second row and second column of the product matrix, which is the dot product of the second row of the first matrix and the second column of the second matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements represented by asterisks (*). The second matrix is a 3x4 matrix, also with all elements represented by asterisks (*). The third matrix is a 5x4 matrix, also with all elements represented by asterisks (*). The second matrix is highlighted with a light red background, and the third matrix is highlighted with a light purple background. The first matrix is highlighted with a light blue background. The second and third matrices are separated by an equals sign (=).

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements represented by asterisks (*). The second matrix is a 3x4 matrix, also with all elements as asterisks, but its fourth column is highlighted with a light red background. An equals sign (=) follows. The third matrix is a 5x4 matrix with all elements as asterisks, but its fourth column is highlighted with a light purple background. This illustrates that the product of the second row of the first matrix and the fourth column of the second matrix results in the fourth column of the third matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with all elements represented by asterisks (*). The second matrix is a 3x4 matrix, also with all elements represented by asterisks. The third matrix is a 5x4 matrix, also with all elements represented by asterisks. An equals sign (=) is placed between the second and third matrices. The first matrix has its third row highlighted in light blue. The second matrix has its first column highlighted in light red. The third matrix has its second row highlighted in light purple. This visualizes the calculation of the element in the third row and second column of the product matrix, which is the dot product of the third row of the first matrix and the second column of the second matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; the third row is highlighted in light blue. The second matrix is a 3x4 matrix with asterisks; the second column is highlighted in light red. An equals sign follows, and then a 5x4 matrix where the element in the third row and second column is highlighted in light purple, representing the result of the dot product of the third row of the first matrix and the second column of the second matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; the third row is highlighted in light blue. The second matrix is a 3x4 matrix with asterisks; the third column is highlighted in light red. An equals sign follows. The third matrix is a 5x4 matrix with asterisks; the element in the third row and third column is highlighted in light purple, representing the dot product of the third row of the first matrix and the third column of the second matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 grid of asterisks with its third row highlighted in light blue. The second matrix is a 3x4 grid of asterisks with its fourth column highlighted in light red. An equals sign follows, and then a 5x4 grid of asterisks where the element at the intersection of the third row and fourth column is highlighted in light purple.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; the fourth row is highlighted in light blue. The second matrix is a 3x4 matrix with asterisks; the first column is highlighted in light red. An equals sign follows. The third matrix is a 5x4 matrix with asterisks; the element in the fourth row and first column is highlighted in light purple, representing the dot product of the fourth row of the first matrix and the first column of the second matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

$$\begin{bmatrix} * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \\ * & * & * \end{bmatrix} \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix}$$

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 grid of asterisks with its fourth row highlighted in light blue. The second matrix is a 3x4 grid of asterisks with its fourth column highlighted in light red. An equals sign follows, and then a 5x4 grid of asterisks with its fourth column highlighted in light purple. This illustrates that the fourth row of the first matrix is multiplied by the fourth column of the second matrix to produce the fourth column of the resulting matrix.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; the bottom row is highlighted with a light blue background. The second matrix is a 3x4 matrix with asterisks; the first column is highlighted with a light red background. An equals sign follows. The third matrix is a 5x4 matrix with asterisks; the bottom-left cell is highlighted with a light purple background.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in all cells; its bottom row is highlighted in light blue. The second matrix is a 3x4 matrix with asterisks in all cells; its second column is highlighted in light red. An equals sign follows, and then a 5x4 matrix with asterisks in all cells; its second column (which is the product of the first row and the second column of the previous matrices) is highlighted in light purple.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

The diagram illustrates the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; the bottom row is highlighted in light blue. The second matrix is a 3x4 matrix with asterisks; the third column is highlighted in light red. An equals sign follows. The third matrix is a 5x4 matrix with asterisks; the element in the bottom row and third column is highlighted in light purple, representing the result of the dot product of the first matrix's bottom row and the second matrix's third column.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Row-Column Rule (Pictorially)

A pictorial representation of the row-column rule for matrix multiplication. It shows three matrices. The first matrix is a 5x3 matrix with asterisks in each cell; the bottom row is highlighted in light blue. The second matrix is a 3x4 matrix with asterisks in each cell; the fourth column is highlighted in light red. An equals sign follows. The third matrix is a 5x4 matrix with asterisks in each cell; the bottom-right element is highlighted in light purple.

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

Matrix Operations

Matrix "Interface"

multiplication

what does AB mean when A and B are matrices?

addition

what does $A + B$ mean when A and B are matrices?

scaling

what does cA mean when A is matrix and c is a real number?

Matrix "Interface"

multiplication

what does AB mean when A and B are matrices?

addition

what does $A + B$ mean when A and B are matrices?

scaling

what does cA mean when A is matrix and c is a real number?

These should be consistent with matrix-vector interface and vector interface

Matrix Addition

$$[\mathbf{a}_1 \quad \dots \quad \mathbf{a}_n] + [\mathbf{b}_1 \dots \quad \mathbf{b}_n] = [(\mathbf{a}_1 + \mathbf{b}_1) \quad \dots \quad (\mathbf{a}_n + \mathbf{b}_n)]$$

Addition is done column-wise (or element-wise)

$$\text{e.g. } \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} (1+2) & (2+3) \\ (3-2) & (4-3) \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & 1 \end{bmatrix}$$

Matrix Addition

$$[\mathbf{a}_1 \quad \dots \quad \mathbf{a}_n] + [\mathbf{b}_1 \dots \quad \mathbf{b}_n] = [(\mathbf{a}_1 + \mathbf{b}_1) \quad \dots \quad (\mathbf{a}_n + \mathbf{b}_n)]$$

Addition is done column-wise (or element-wise)

$$\text{e.g. } \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 3 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} (1+2) & (2+3) \\ (3-2) & (4-3) \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 1 & 1 \end{bmatrix}$$

This is exactly the same as vector addition, but for matrices

Matrix Addition and Scaling

$$c [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] = [c\mathbf{a}_1 \quad c\mathbf{a}_2 \quad \dots \quad c\mathbf{a}_n]$$

Scaling and adding happen element-wise (or, equivalently, column-wise)

$$\text{e.g. } 2 \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 2(1) & 2(2) \\ 2(-1) & 2(3) \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ -2 & 6 \end{bmatrix}$$

Matrix Addition and Scaling

$$c [\mathbf{a}_1 \quad \mathbf{a}_2 \quad \dots \quad \mathbf{a}_n] = [c\mathbf{a}_1 \quad c\mathbf{a}_2 \quad \dots \quad c\mathbf{a}_n]$$

Scaling and adding happen element-wise (or, equivalently, column-wise)

$$\text{e.g. } 2 \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 2(1) & 2(2) \\ 2(-1) & 2(3) \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ -2 & 6 \end{bmatrix}$$

This is exactly the same as vector scaling, but for matrices

Algebraic Properties (Addition and Scaling)

$$A + B = B + A$$

$$(A + B) + C = A + (B + C)$$

$$A + 0 = A$$

$$r(A + B) = rA + rB$$

$$(r + s)A = rA + sA$$

$$r(sA) = (rs)A$$

In these properties A , B , and C are matrices of the same size and r and s are scalars ($\mathbb{R}$)

We need to know/memorize these

Algebraic Properties (Addition and Scaling)

$$A(BC) = (AB)C$$

$$A(B + C) = AB + AC$$

$$(B + C)A = BA + CA$$

$$r(AB) = (rA)B = A(rB)$$

$$I_m A = A = A I_n$$

In these properties A , B , and C are matrices of the appropriate size so that everything is defined, and r is a scalar

We need to know/memorize these

Matrix Multiplication is not Commutative

Important. AB may not be the same as BA

(it may not even be defined)

More Matrix Operations

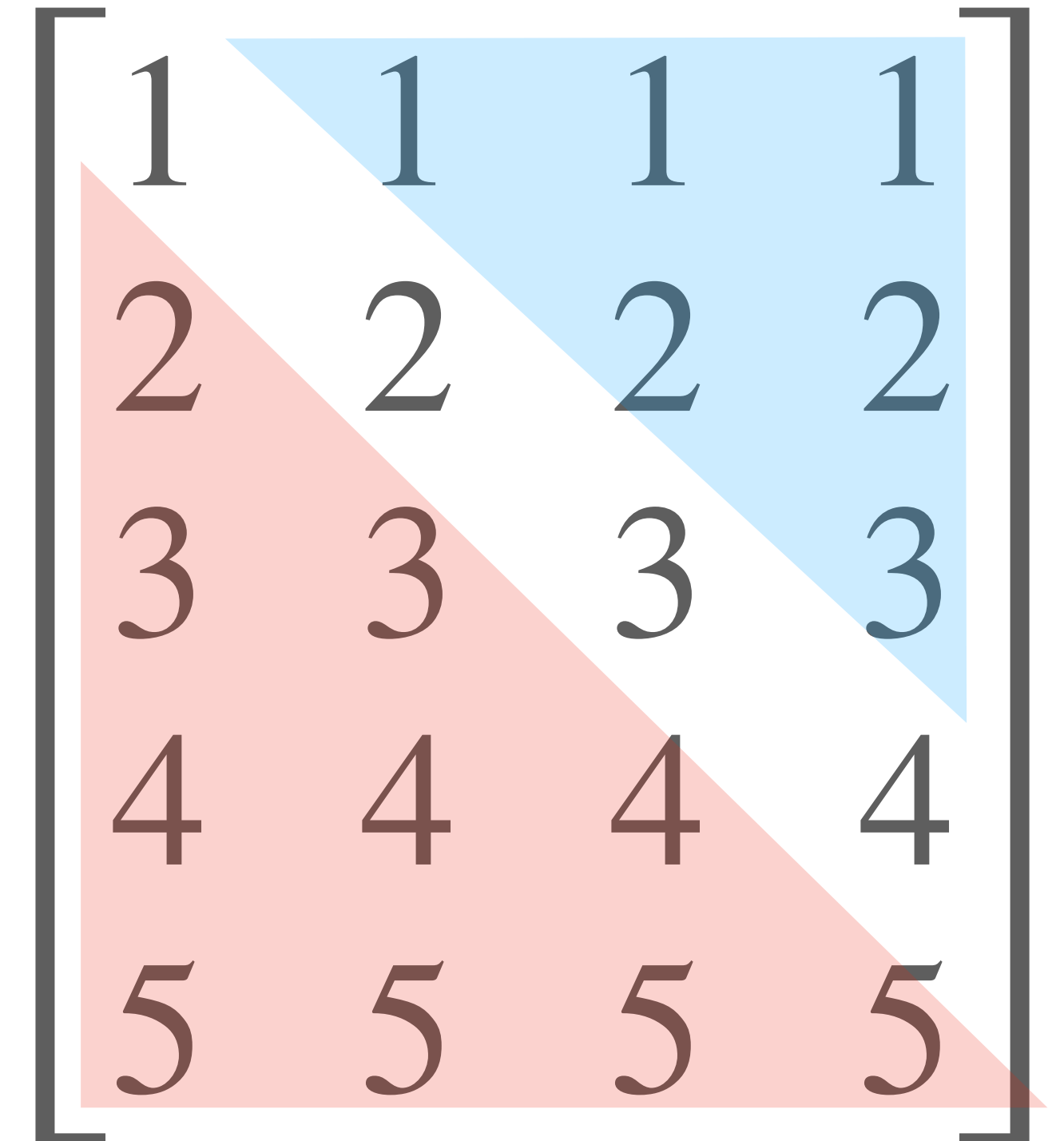
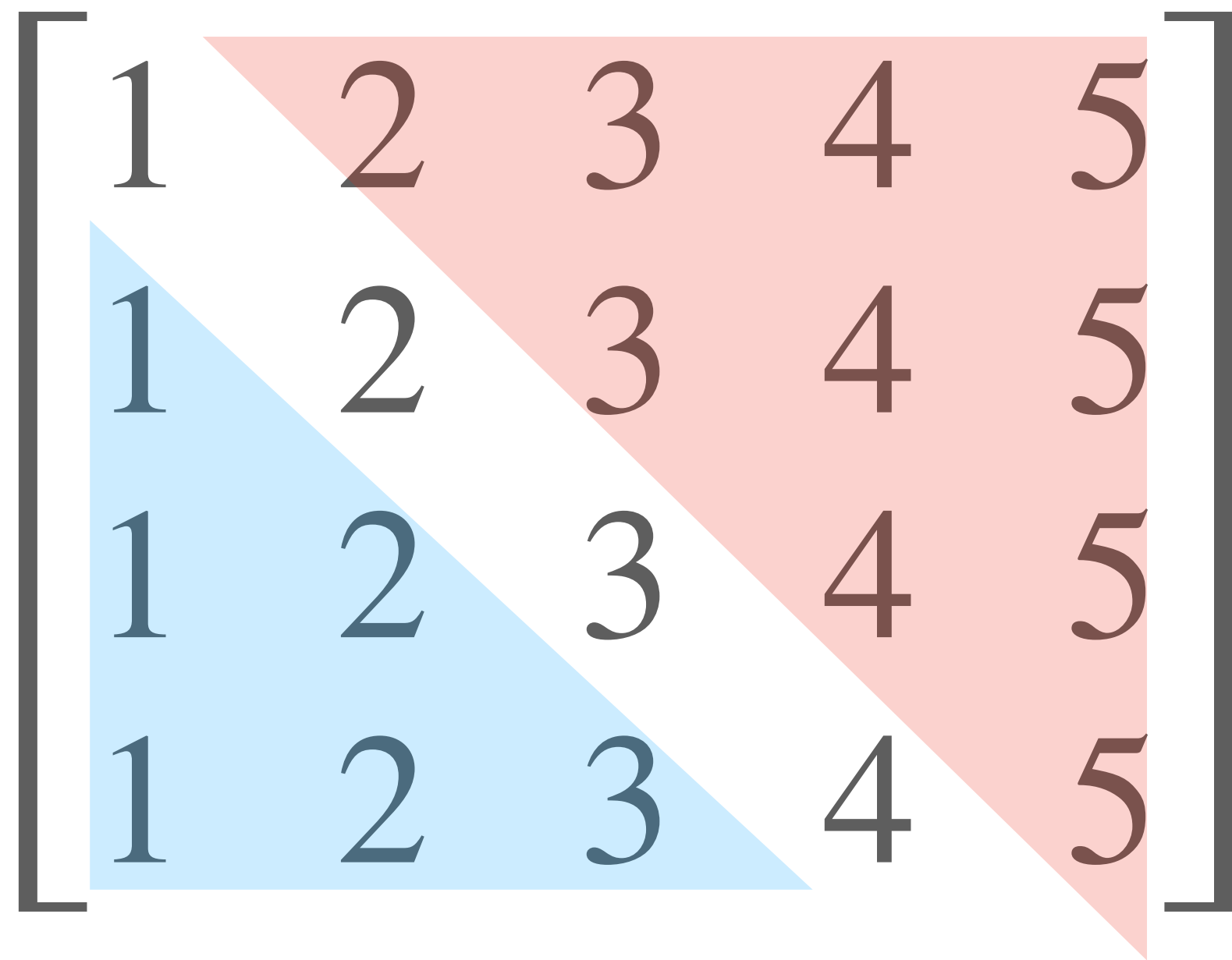
Transpose (Pictorially)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 \end{bmatrix}$$



$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 5 & 5 & 5 & 5 \end{bmatrix}$$

Transpose (Pictorially)



Transpose

Definition. For a $m \times n$ matrix A , the **transpose** of A , written A^T , is the $n \times m$ matrix such that

$$(A^T)_{ij} = A_{ji}$$

Example.

$$\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Algebraic Properties (Transpose)

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(cA)^T = cA^T \text{ (where } c \text{ is a scalar)}$$

$$(AB)^T = B^T A^T$$

Algebraic Properties (Transpose)

$$(A^T)^T = A$$

$$(A + B)^T = A^T + B^T$$

$$(cA)^T = cA^T \text{ (where } c \text{ is a scalar)}$$

$$(AB)^T = B^T A^T \text{ Important: the order reverses!}$$

Matrix Powers

Matrix Powers

Obs. If A is an $n \times n$ matrix, then the product AA is defined

Matrix Powers

Obs. If A is an $n \times n$ matrix, then the product AA is defined

Def. For $A \in \mathbb{R}^{n \times n}$, we write A^k for the k -fold product of A with itself

Matrix Powers

Obs. If A is an $n \times n$ matrix, then the product AA is defined

Def. For $A \in \mathbb{R}^{n \times n}$, we write A^k for the k -fold product of A with itself

What should A^0 be? (we want $A^0 A^k = A^{0+k} = A^k$)

Final Warnings about Matrix Multiplication

Final Warnings about Matrix Multiplication

1. AB is not necessarily equal to BA , even if both are defined

Final Warnings about Matrix Multiplication

1. AB is not necessarily equal to BA , even if both are defined
2. If $AB = AC$ then it is not necessary that $B = C$

Final Warnings about Matrix Multiplication

1. AB is not necessarily equal to BA , even if both are defined
2. If $AB = AC$ then it is not necessary that $B = C$
3. If $AB = 0$ (the zero matrix) it is not necessarily the case that $A = 0$ or $B = 0$

Summary

We can reason about matrix equations by reasoning directly about properties of linear transformations

Matrix multiplication coincides with composition of linear transformations

There is an algebra of matrices which is consistent with the algebra of vectors